Module B4 — Understanding Processes

4.1 Waves and Wave Properties

Wave: A disturbance that transfers energy from one place to another without a net transfer of matter. Waves can be described by their amplitude, wavelength, frequency, and speed.

Transverse wave: A wave in which the oscillation is perpendicular to the direction of energy transfer. Examples: electromagnetic waves, waves on a string, S-waves.

Longitudinal wave: A wave in which the oscillation is parallel to the direction of energy transfer. Examples: sound waves, P-waves, compression waves in a spring.

Wave Equation

$$v = f\lambda$$

where v is the wave speed (m s^{-1}), f is the frequency (Hz), and λ is the wavelength (m).

Key Wave Quantities

Quantity	Symbol	Definition	Unit
Amplitude	A	Maximum displacement from the equilibrium position	m
Wavelength	λ	Minimum distance between two points oscillating in phase	m
Frequency	f	Number of complete oscillations per unit time	Hz (s^{-1})

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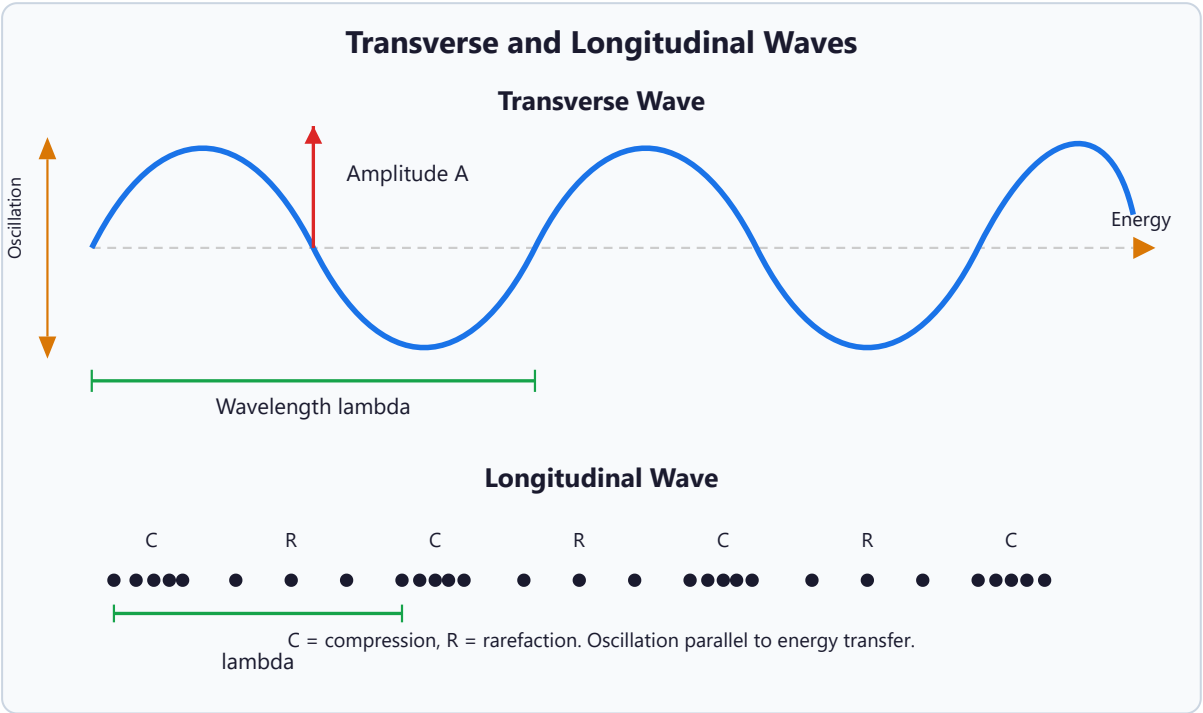
Period	T	Time for one complete oscillation; T = 1/f	s
Phase difference	φ	The fraction of a cycle by which one wave leads or lags another	radians or degrees



Phase difference: Two points oscillating in step are in phase (phase difference = 0 or 2π). Two points oscillating in opposite directions are in antiphase (phase difference = π or 180°). Phase difference can be expressed as $\varphi = (2\pi\Delta x)/\lambda$ where Δx is the path difference.

Superposition and Interference

Principle of superposition: When two or more waves meet at a point, the resultant displacement is the vector sum of the individual displacements at that instant.

Interference occurs when two coherent waves overlap. Coherent sources have the same frequency and a constant phase difference.

- **Constructive interference:** Occurs when waves arrive in phase (path difference = nλ). The resultant amplitude is the sum of the individual amplitudes.

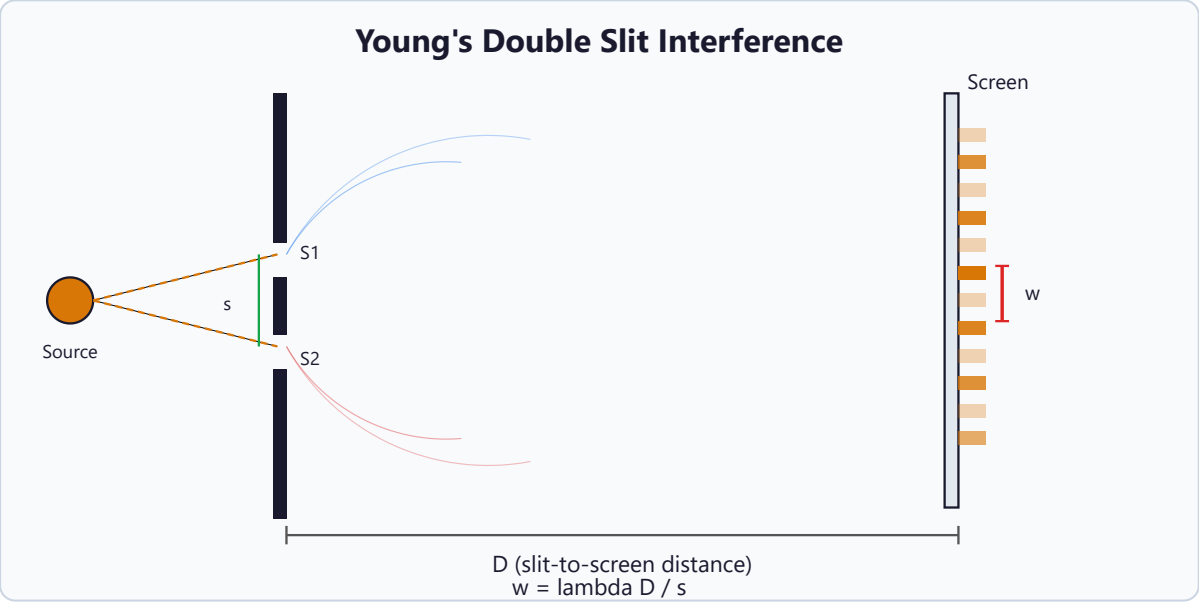
- **Destructive interference:** Occurs when waves arrive in antiphase (path difference = (n + ½)λ). The resultant amplitude is the difference of the individual amplitudes.

Young's Double Slit Experiment

Young's experiment demonstrates the wave nature of light by producing an interference pattern of bright and dark fringes on a screen. Two narrow slits act as coherent sources.

$$\text{Fringe spacing: } w = \lambda D / s$$

where w is the fringe spacing, λ is the wavelength, D is the distance from slits to screen, and s is the slit separation. This equation only applies when D is much greater than s.



Exam Tip: Young's double slit experiment is evidence for the wave nature of light. Be prepared to describe how the experiment is set up, what is observed, and how changing λ, D, or s affects the fringe pattern. Increasing λ or D increases fringe spacing; increasing s decreases it.

Stationary (Standing) Waves

Stationary wave: A wave pattern formed by the superposition of two progressive waves of the same frequency and amplitude travelling in opposite directions. Stationary waves have nodes (points of

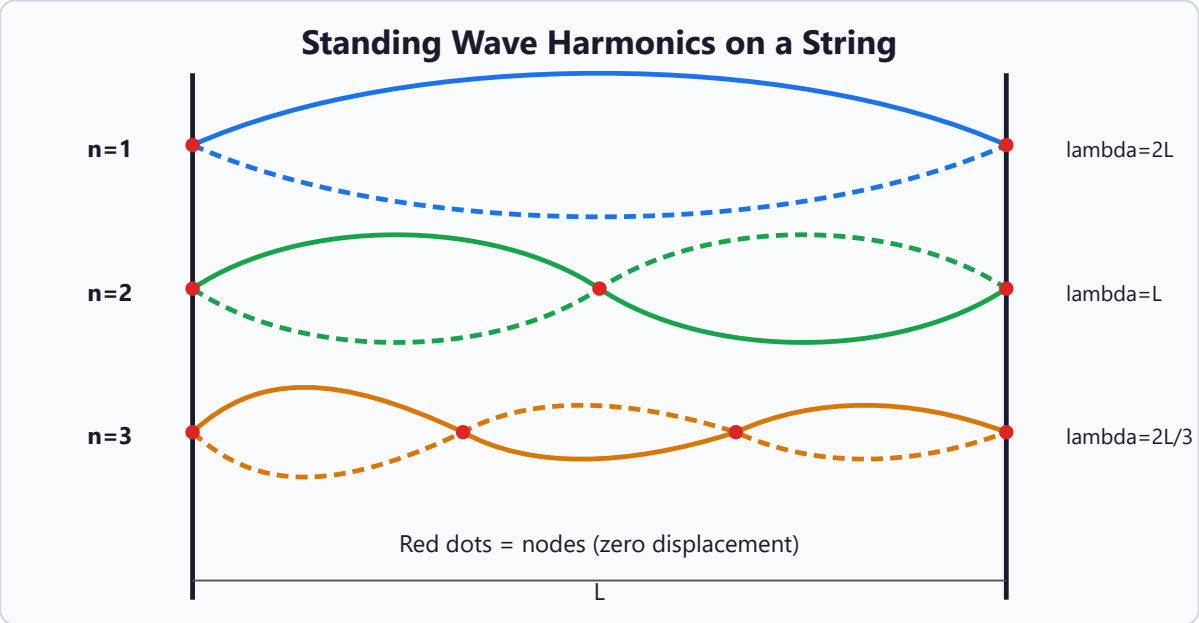
zero displacement) and antinodes (points of maximum displacement). Unlike progressive waves, energy is not transferred along a stationary wave.

Stationary waves on a string have nodes at the fixed ends. The fundamental mode (first harmonic) has one antinode and two nodes. The frequency of the nth harmonic is:

$$f_n = n \times f_1 = n / (2L) \times v$$

where L is the length of the string and v is the wave speed on the string.

Harmonic	Nodes	Antinodes	Wavelength
1st (fundamental)	2	1	λ = 2L
2nd	3	2	λ = L
3rd	4	3	λ = 2L/3
nth	n + 1	n	λ = 2L/n



4.2 Quantum Behaviour

The Photon Model

Photon: A quantum (discrete packet) of electromagnetic radiation. The energy of a photon is proportional to its frequency and inversely proportional to its wavelength.

$$E = hf = hc / \lambda$$

where h is the Planck constant (6.63×10^{-34} J s), f is the frequency, c is the speed of light, and λ is the wavelength.

The electronvolt (eV) is a convenient unit for photon energies: 1 eV = 1.60×10^{-19} J. Visible light photons have energies of roughly 1.8 eV (red) to 3.1 eV (violet).

The Photoelectric Effect

Photoelectric effect: The emission of electrons from a metal surface when electromagnetic radiation of sufficiently high frequency is incident upon it. The photoelectric effect provides evidence for the particle nature of light.

Key observations of the photoelectric effect that wave theory cannot explain:

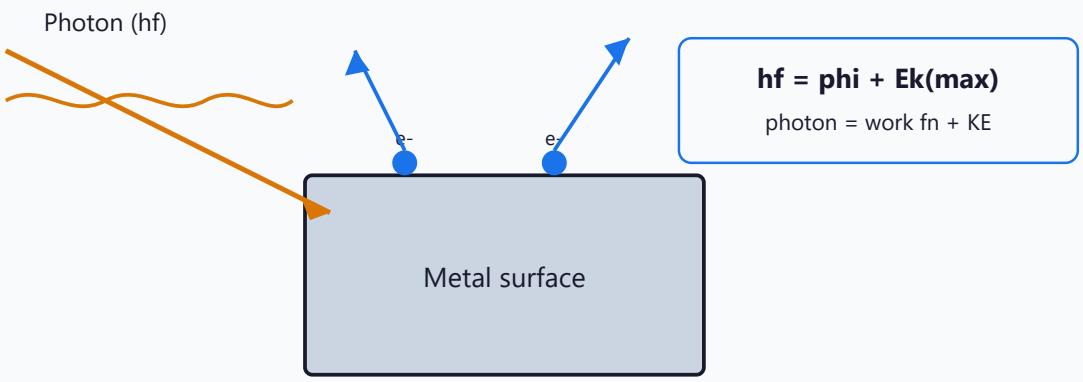
- Electrons are emitted only when the frequency exceeds a threshold frequency f_0, regardless of intensity
- Increasing intensity increases the number of electrons emitted (photocurrent) but not their maximum kinetic energy
- Increasing frequency above the threshold increases the maximum kinetic energy of emitted electrons
- Emission is instantaneous, with no time delay even at low intensities

$$\text{Einstein's photoelectric equation: } hf = \varphi + E_{k(max)}$$

where φ is the work function of the metal (the minimum energy needed to release an electron from the surface) and $E_{k(max)}$ is the maximum kinetic energy of the emitted photoelectrons.

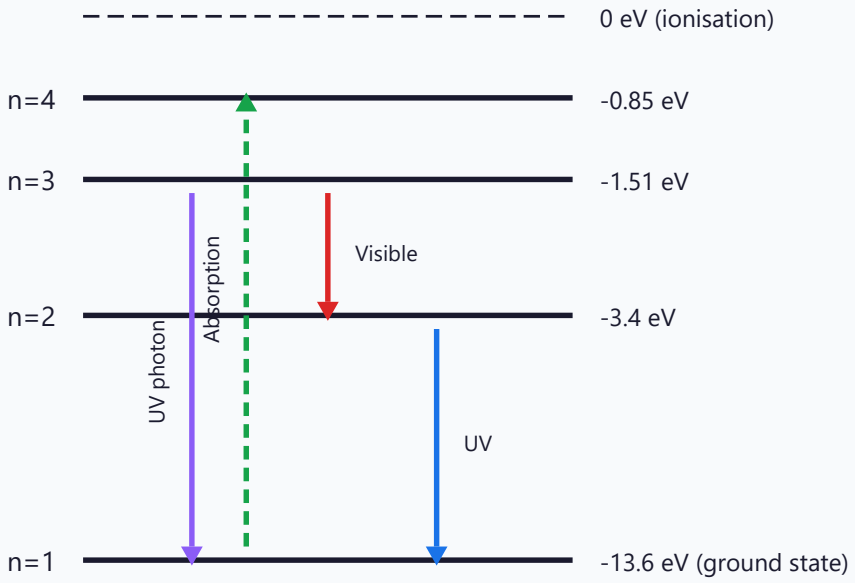
$$\text{Threshold frequency: } f_0 = \varphi / h$$

The Photoelectric Effect



Below threshold frequency f0: NO electrons emitted
Above f0: increasing f increases max KE
Increasing intensity increases number of electrons, not max KE

Energy Level Diagram with Transitions



E(photon) = hf = E(upper) - E(lower)

Exam Tip: The photoelectric effect is one of the most commonly examined topics. Remember that it is evidence for the particle nature of light. A single photon interacts with a single electron — it is an all-or-nothing process. Intensity affects the number of photons per second, not the energy per photon.

Wave-Particle Duality

Wave-particle duality: The concept that all matter and radiation exhibit both wave-like and particle-like properties. Light shows wave behaviour (interference, diffraction) and particle behaviour (photoelectric effect). Electrons show particle behaviour (tracks in cloud chambers) and wave behaviour (electron diffraction).

De Broglie proposed that all moving particles have an associated wavelength:

$$\text{de Broglie wavelength: } \lambda = h / p = h / (mv)$$

where p is the momentum, m is the mass, and v is the velocity. For everyday objects, the de Broglie wavelength is incredibly small and wave effects are undetectable. For electrons accelerated through a potential difference, the wavelength can be comparable to atomic spacings, allowing electron diffraction to be observed.

Electron Diffraction

When electrons are accelerated through a thin graphite film, they produce a diffraction pattern of concentric rings on a fluorescent screen. This provides direct evidence that electrons have wave-like properties. Increasing the accelerating voltage increases the electron's momentum, decreasing the de Broglie wavelength, which reduces the diameter of the diffraction rings.

Quantum Behaviour of Photons

OCR B takes a distinctive approach to quantum behaviour, emphasising the probabilistic nature of quantum mechanics. Key ideas include:

- A photon does not follow a definite path. Instead, we must consider all possible paths from source to detector.
- Each path has an associated amplitude (represented by a rotating "phasor" arrow).
- The probability of a photon arriving at a particular point is proportional to the square of the resultant amplitude found by adding all the phasor arrows.
- This "many paths" approach naturally explains reflection, refraction, diffraction, and interference without needing separate rules for wave and particle behaviour.

Exam Tip: The OCR B "many paths" approach to quantum behaviour is distinctive to this specification. Be comfortable explaining how phasor arrows from different paths combine, and how

the squared magnitude of the resultant gives the detection probability. This is key to understanding why light appears to travel in straight lines (constructive interference along direct paths) and why diffraction occurs (fewer paths to cancel at the edges).

4.3 Space, Time, and Motion

Frames of Reference

Frame of reference: A coordinate system relative to which the position and motion of objects can be measured. An inertial frame is one that is not accelerating — Newton's laws hold in their simplest form in inertial frames.

Special Relativity — Basic Ideas

Einstein's special theory of relativity is built on two postulates:

1. The laws of physics are the same in all inertial frames of reference
2. The speed of light in a vacuum (c) is the same for all observers, regardless of the motion of the source or observer

These seemingly simple postulates lead to profound consequences when objects move at speeds comparable to c:

Time Dilation

A clock moving relative to an observer runs more slowly than a clock at rest relative to that observer.

$$t = t_0 / \sqrt{(1 - v^2/c^2)} = \gamma t_0$$

where t_0 is the proper time (measured in the frame where the clock is at rest), t is the dilated time (measured by the observer relative to whom the clock is moving), and γ is the Lorentz factor.

$$\text{Lorentz factor: } \gamma = 1 / \sqrt{(1 - v^2/c^2)}$$

Length Contraction

An object moving relative to an observer is measured to be shorter in the direction of motion than when at rest.

$$L = L_0 / \gamma = L_0\sqrt{(1 - v^2/c^2)}$$

where L_0 is the proper length (measured in the frame where the object is at rest) and L is the contracted length measured by the moving observer.

Proper time: The time interval between two events measured by an observer for whom the events occur at the same place.

Proper length: The length of an object measured by an observer at rest relative to the object.

Mass-Energy Equivalence

$$E = mc^2$$

This famous equation shows that mass and energy are equivalent and interchangeable. A small amount of mass corresponds to an enormous amount of energy. This is observed in nuclear reactions (fission and fusion) where the mass of the products is less than the mass of the reactants, and the difference (mass defect) is released as energy.

Exam Tip: At A-Level, relativity questions typically involve straightforward substitution into the time dilation or length contraction formulae. Always identify which frame measures the proper time or proper length. The proper time is always the shorter time; the proper length is always the longer length.

Evidence for Special Relativity

Experimental evidence supporting special relativity includes:

- **Muon decay:** Muons created in the upper atmosphere by cosmic rays travel much further than expected given their short half-life. Time dilation explains their extended lifetime as observed from Earth.

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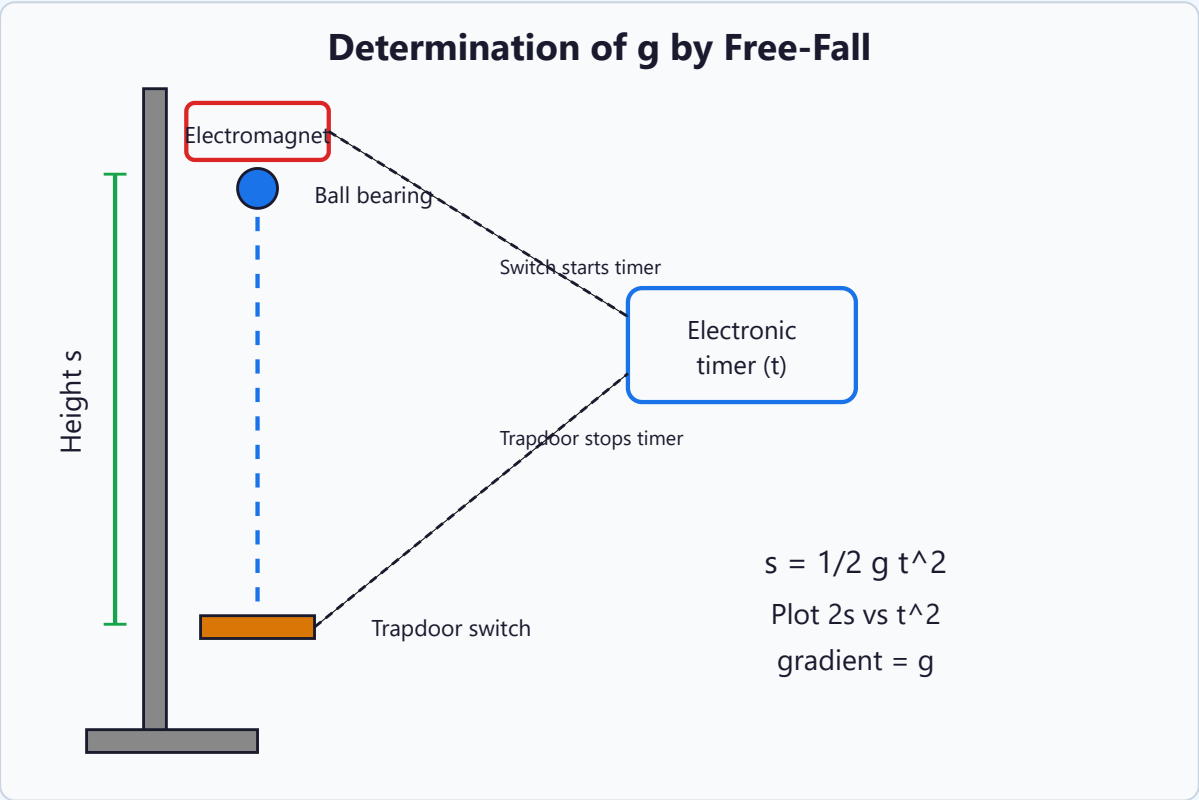
- **Particle accelerators:** Particles approaching the speed of light require increasingly more energy to accelerate, consistent with relativistic mass increase.
- **GPS satellites:** Corrections for both special and general relativistic effects are essential for accurate GPS positioning.

Practical Activity: Determination of g by Free-Fall



Apparatus

- Electromagnet (to hold and release the ball bearing)
- Steel ball bearing
- Trapdoor switch (at the bottom, connected to timer)
- Electronic timer (millisecond resolution)
- Metre ruler or measuring tape (to measure height s)
- Clamp stand

Method

1. Set up the electromagnet at a measured height s above the trapdoor.
2. Hold the ball bearing with the electromagnet. Switch off the electromagnet to release the ball — this simultaneously starts the electronic timer.
3. The ball falls and hits the trapdoor, which opens and stops the timer. Record time t.
4. Repeat 3 times at this height and take the mean time.
5. Change the height s (e.g., from 0.2 m to 1.0 m in 0.1 m steps) and repeat.
6. Plot a graph of 2s (y-axis) against t^2 (x-axis). The gradient equals g.

Safety

- Place a cushion or sand tray beneath the trapdoor to catch the ball bearing
- Ensure the clamp stand is stable and will not topple
- Keep feet away from the drop zone

Analysis

From $s = (1/2)gt^2$, rearranging gives $2s = gt^2$. Plotting 2s against t^2 gives a straight line through the origin with gradient = g. The accepted value is 9.81 m s^{-2}.

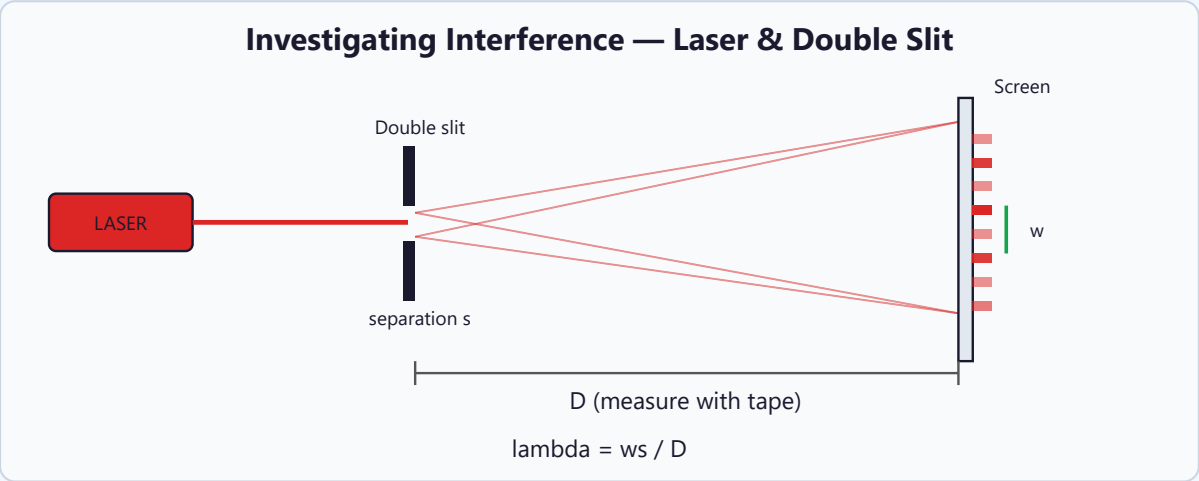
Sources of Error & Improvements

Source of Error	Type	Improvement
Delay between electromagnet switching off and ball releasing (residual magnetism)	Systematic (makes t too large, g too small)	Use a larger range of heights; the y-intercept of the 2s vs t^2 graph accounts for the systematic delay
Measuring the height from the bottom of the ball to the trapdoor	Random	Use a set square to accurately position the ruler; measure to the bottom of the ball
Air resistance on the ball	Systematic (makes g appear smaller)	Use a small, dense ball bearing; keep heights less than ~2 m

Exam Tip: If using light gates instead, measure the time for the ball to fall between two gates. With two light gates, you can also measure the velocity at each gate and use $v^2 = u^2 + 2as$

to find g. The advantage of light gates is that they eliminate human reaction time errors.

Practical Activity: Investigating Interference and Diffraction



Apparatus

- Laser (monochromatic, coherent light source — e.g., red helium-neon, lambda ~ 633 nm)
- Double slit (known slit separation s, typically 0.1-0.5 mm)
- White screen or wall
- Metre ruler or tape measure (to measure D and fringe spacing)
- Darkened room

Method

1. Set up the laser to shine through the double slit onto a screen at distance D (at least 1–2 m away for clear fringes).
2. Observe the interference pattern: equally spaced bright and dark fringes.
3. Measure the distance across several fringes (e.g., across 10 bright fringes) and divide by the number of fringe spacings to get the mean fringe spacing w.
4. Measure the distance D from the double slit to the screen.
5. Calculate the wavelength: lambda = ws / D.

6. Repeat with the screen at different distances D and check that lambda is consistent.

Safety

- NEVER look directly into the laser beam or its reflections — risk of permanent eye damage
- Display a laser safety warning sign
- Remove reflective objects (watches, jewellery) from the beam path
- Switch off the laser when not in use

Analysis

From w = lambda D / s, rearranging gives lambda = ws / D. Alternatively, plot w (y-axis) against D (x-axis) — the gradient equals lambda / s, so lambda = gradient x s. The result should agree with the known wavelength of the laser.

Sources of Error & Improvements

Source of Error	Type	Improvement
Difficulty measuring small fringe spacing	Random	Increase D to spread fringes further apart; measure across many fringes and divide
Laser beam not perpendicular to the slits	Systematic	Align the laser carefully; check the central maximum is at the centre of the screen
Slit separation s not accurately known	Systematic	Use manufacturer's stated value or measure with a travelling microscope

Exam Tip: If asked why a laser is used rather than a white light source: a laser produces coherent, monochromatic light, giving clearly defined and evenly spaced fringes. White light produces fringes with coloured edges that quickly merge due to the range of wavelengths.

Worked Example: In a Young's double slit experiment, the slit separation is 0.50 mm, the screen is 2.0 m away, and the fringe spacing is measured as 2.5 mm. Calculate the wavelength of the light.

$w = \lambda D / s$, so $\lambda = ws / D$

$\lambda = (2.5 \times 10^{-3} \times 0.50 \times 10^{-3}) / 2.0$

$\lambda = 1.25 \times 10^{-6} / 2.0 =$ **6.25×10^{-7} m = 625 nm**

This is orange-red light, consistent with the visible spectrum (400–700 nm).

Worked Example: Calculate the critical angle for light travelling from glass (n = 1.50) into air (n = 1.00).

$\sin \theta_c = n_2 / n_1 = 1.00 / 1.50 = 0.667$

$\theta_c = \sin^{-1}(0.667) =$ **41.8°**

Any light hitting the glass-air boundary at an angle greater than 41.8° (measured from the normal) will be totally internally reflected. This is how optical fibres and prisms in binoculars work.